MATH 4920: Undergraduate Research Projects

Manitoba eXperimental Mathematics Laboratory

Chaotic Transient Dynamics in Ecological and Physical Systems

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1 Introduction

Historically, the study of dynamical systems through mathematical modelling has focused on long-term asymptotic behavior, such as stable equilibria or limit cycles. While this research offers important insights into the eventual fate of a system, it neglects the dynamics that occur before the system settles. These 'transient dynamics' are non-trivial and in many systems, especially those in ecology, persist for several generations. Many real-world phenomena, such as the sudden onset and persistence of severe algal blooms, are effectively in this transient state for the long term, and as such the study of these transient dynamics can be more relevant and useful. Understanding the behavior, duration, and severity of these transients is crucial for making predictions about the future state of the system.

This project studies the Hastings-Powell (HP) model [1] to investigate these dynamics - a continuous-time, three-species trophic system consisting of a resource, prey, and predator. The HP model is a canonical system in mathematical ecology, and is structurally one of the simplest biological continuous-time models to exhibit chaos. This is grounded in the Poincaré-Bendixson theorem, which dictates that continuous-time differential equations require at least three dimensions (in this case, three interacting species in a food chain) to exhibit chaotic behavior [2].

Conceptually, chaos describes a system that is entirely deterministic yet highly sensitive to initial conditions, rendering long-term forecasting essentially useless [3]. This is particularly important in the context of transient dynamics; a chaotic system is trapped in a (perhaps prolonged) state of irregularity instead of predictable convergence to a stable steady state. Because of this, the 'temporary' transient phase essentially becomes the permanent observable reality of the system. In the HP model, this chaotic behavior arises from the non-linear saturation of the predators, that is, the biological constraint that consumption rates plateau as the prey density increases. While this saturation does not guarantee chaotic behavior on its own, under certain parameter regimes it acts as a critical mechanism that allows populations to escape regulation and enter chaotic transient states [4, 1].

A key objective of this project is to approach this problem in a cross-disciplinary manner by investigating the mathematical isomorphism between this ecological saturation and non-linear phenomena in laser physics. Specifically, the satiation of a predator in the HP model is analogous to the bleaching of a saturable absorber in a Q-switched laser. These systems

both exhibit bursting behaviors: population booms in ecology and large pulses in optics driven by mathematically similar accumulation and release mechanisms. By leveraging tools such as the Finite Time Lyapunov Exponent (FTLE) from chaos theory, I aim to quantify the predictability of these transient events and study them in detail. In this report, Section 2 outlines the background models, Section 3 covers the linear stability analysis, Section 4 outlines the computational methodologies alongside the numerical results, and Section 5 presents the concluding remarks and future directions.

2 Theoretical Background

2.1 Project Objectives and Individual Contributions

The primary focus of this project is to study the mechanisms that drive transient dynamics in chaotic systems, instead of considering just the traditional long-term asymptotic behavior. Specifically, this project aims to:

- Investigate the mathematical isomorphism between the ecological Hastings-Powell model and the optical Yamada laser model.
- Demonstrate that the transient accumulation-and-release mechanisms in both systems are driven by equivalent saturation mechanics (handling time in ecology and optical bleaching in physics).
- Map the route to chaos via bifurcation analysis to show the transition from stable equilibria to chaotic dynamics.
- Quantify the predictability and duration of transient stability using Finite Time Lyapunov Exponents (FTLEs)

2.2 The Hastings-Powell Model

In this section, we introduce the theoretical framework of the Hastings-Powell (HP) model. The HP system models a continuous-time trophic food chain consisting of three distinct populations: a foundational resource (X), a prey species (Y), and an apex predator (Z).

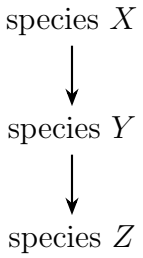


Figure 1: Diagram of the three-species trophic chain modeled by the Hastings-Powell system. Arrows indicate the flow of biomass (predation).

The dynamics of this 3-species food chain as shown in Figure 1 can be expressed as a system

of 3 ODEs as shown below.

$$\begin{aligned}\frac{dX}{dT} &= R_0 X\left(1-\frac{X}{K_0}\right) - C_1 F_1(X)Y \\ \frac{dY}{dT} &= F_1(X)Y - F_2(Y)Z - D_1 Y \\ \frac{dZ}{dT} &= C_2 F_2(Y)Z - D_2 Z\end{aligned} \tag{1}$$

with the Holling Type II functional response defined as

$$F_i(U) = \frac{A_i U}{B_i + U} \quad \text{for } i = 1, 2. \tag{2}$$

The parameters governing these equations are defined in Table 1

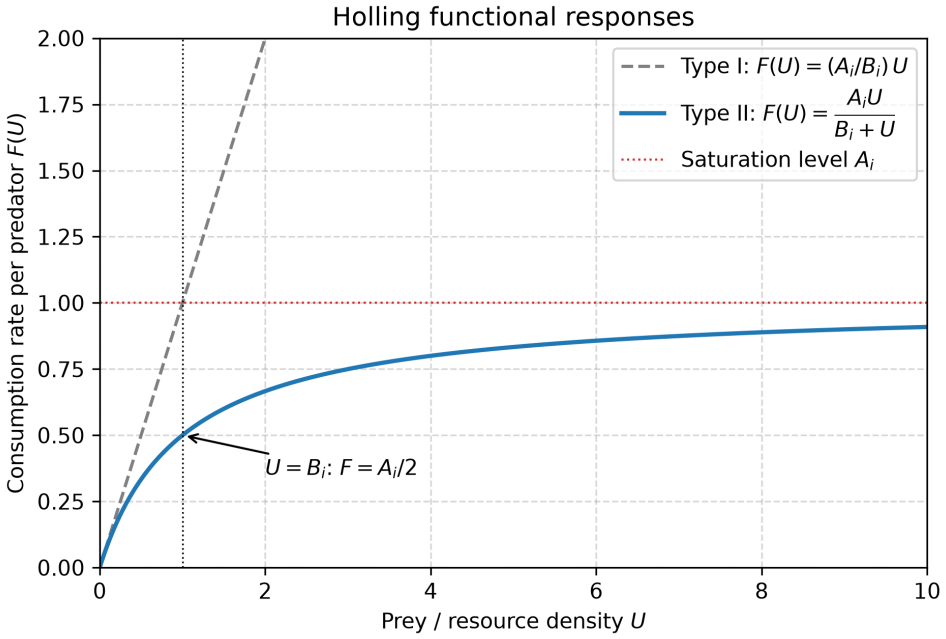


Figure 2: Holling Functional Responses: The linear Type I response (dashed) implies unlimited consumption, whereas the Type II response (solid) saturates at a maximum rate A_i, representing the predator's handling time constraints.

As shown in Figure 2, the dashed line represents a linear (Type I) functional response, while the solid line represents the Holling Type II response used in the HP model. In the linear Type I case, the consumer eats in proportion to its target's density. It exerts constant control over the prey; in such a system, the lower trophic level can never break out or bloom, as they are consumed just as fast as they reproduce. In contrast, the Holling Type II curve behaves linearly at low resource densities ($X \to 0$), where the predator consumes prey easily. However, at high prey densities ($X \to \infty$), the predation rate flattens to a maximum value of A_i. This plateau represents the biological constraint of the predator's handling time.

The disparity between the two curves illustrates the loss of control from the top-down: once the predator is saturated, it cannot keep up with reproduction of the prey, allowing prey populations to escape regulation and form transient outbreaks.

Parameter	Definition
X, Y, Z	Populations of Resource, Prey, and Predator
R_0	Intrinsic growth rate of the resource
K_0	Carrying capacity of the resource
C_1, C_2	Conversion efficiencies
D_1, D_2	Mortality rates of Prey and predator
A_i	Maximum predation/consumption rate
B_i	Half-saturation constant for Type II response

Table 1: Parameters in the Hastings-Powell Model

This raw system of equations (1) has a total of 10 parameters, which can prove to be difficult or intensive to analyze mathematically. Following the reduction procedure originally outlined by Hastings and Powell [1], we define a new set of dimensionless variables:

$$\begin{aligned} t &= R_0 T \\ x &= \frac{X}{K_0} \\ y &= \frac{C_1 Y}{K_0} \\ z &= \frac{C_1 Z}{C_2 K_0} \end{aligned} \tag{3}$$

Substituting these new parameters into (1) we obtain the non-dimensionalized system:

$$\begin{aligned} \frac{dx}{dt} &= x(1-x) - f_1(x)y \\ \frac{dy}{dt} &= f_1(x)y - f_2(y)z - d_1 y \\ \frac{dz}{dt} &= f_2(y)z - d_2 z \end{aligned} \tag{4}$$

where $f_i(u) = \frac{a_i u}{1+b_i u}$ for $i = 1, 2$ represents the non-dimensionalized Holling Type II functional response. The simplified parameters are given in terms of the original dimensional constants as: $a_1 = \frac{A_1 K_0}{B_1 R_0}$, $b_1 = \frac{K_0}{B_1}$, $a_2 = \frac{A_2 C_2 K_0}{B_2 C_1 R_0}$, $b_2 = \frac{K_0}{B_2 C_1}$, $d_1 = \frac{D_1}{R_0}$, and $d_2 = \frac{D_2}{R_0}$.

2.3 Isomorphism to Q-Switched Laser Dynamics

Transient phenomena, such as the explosive onset and subsequent crash of ecological outbreaks, are notoriously difficult to forecast using standard ecological analysis. However, the chaotic bursting exhibited by the Hastings-Powell model is not exclusively a biological phenomenon; it belongs to a much broader universal class of non-linear accumulation-and-release mechanisms. In laser physics, researchers have spent decades developing robust mathematical frameworks to analyze mathematically analogous bursting behaviors, specifically the optical pulsing of Q-switched lasers. The structural isomorphism between these two models allows for the application of established theorems on laser instability in the field of ecology.

It is helpful to first understand the physical mechanics of a Q-switched laser. The core of the system is the "gain medium"—a physical material, such as a specially doped crystal or a gas, filled with atoms capable of absorbing and storing energy. This medium acts as the foundational environment, entirely analogous to the base biological resource (such as vegetation or nutrients) in a food chain. To "feed" this medium, an "optical pump" (typically a bright lamp or any other external continuous light source) is used to continuously inject energy into the system. This

pump energetically excites the atoms within the gain medium, driving a steady accumulation of stored energy in the exact same way that sunlight drives the intrinsic growth rate of a biological resource.

This actively pumped gain medium is placed inside a laser cavity between two mirrors: one fully reflective mirror that bounces all light back into the system and one partially transparent mirror that allows light to escape only when it reaches a critical intensity. However, to regulate this growing energy, a "saturable absorber" is placed inside the cavity. Physically, this is often another specifically doped crystal or a specialized dye that acts as an optical barrier. At low light intensities, its unexcited atoms easily absorb the passing photons, trapping the light and preventing it from escaping (analogous to a predator regulating a growing prey population). As the trapped light intensifies, the sheer volume of photons eventually excites all the available atoms in the absorber. Once these atoms are fully excited, they can no longer absorb light and are overwhelmed, making the barrier suddenly transparent (bleaching). This releases all the accumulated energy in one massive, violent pulse. This physical choke-point and subsequent sudden release is dynamically identical to an ecological system where a predator becomes saturated by a growing prey population. Once the predator's handling time is maxed out, it loses top-down control, allowing the prey population to experience a rapid outbreak. Both systems are driven by this identical mathematical mechanism of steady accumulation followed by a rapid saturating release.

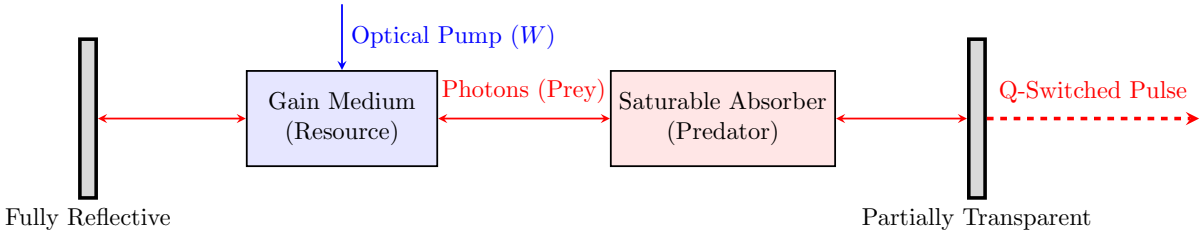


Figure 3: Schematic of a Q-switched laser cavity. The optical pump builds energy in the gain medium, while the saturable absorber prevents emission until it becomes saturated (bleached), resulting in a massive release of photons.

In laser physics, the dynamics of a cavity containing a saturable absorber can be modelled using a three-variable extension of the Statz-deMars equations. The parallelism between the HP model with three variables Resource (X), Prey (Y), and Predator (Z) can be seen here with three variables Gain (n_g), Photon Density (q), and Saturable Absorber (n_a):

$$\begin{aligned} \frac{dn_g}{dt} &= W - \gamma_g n_g - B_g n_g q \\ \frac{dq}{dt} &= (B_g n_g - B_a n_a - \gamma_c)\, q \\ \frac{dn_a}{dt} &= \gamma_a (n_{a0} - n_a) - B_a n_a q \end{aligned} \tag{5}$$

Here, the variables n_g, q, and n_a represent volumetric number densities (cm^{-3}), while the rates (γ and W) operate in inverse seconds (s^{-1}). The optical gain n_g represents population inversion. While the physical gain density is strictly positive, its rate of change (dn_g/dt) is not strictly increasing; it constantly accumulates energy from the optical pump rate W, but is simultaneously depleted by stimulated emission ($-B_g n_g q$). The photons q act as the prey, multiplying rapidly by consuming this gain. Finally, while the saturable absorber is a fixed physical component within the cavity, the variable n_a does not represent its physical mass but rather the population density of its unexcited molecules. As these molecules absorb photons,

$(-B_a n_a q)$, they become energetically excited and transparent (bleached), effectively losing their regulatory grip on the light. They continually attempt to relax back to their absorptive ground state n_{a0} at a rate γ_a.

Similar to how the HP system originally had a large and difficult parameter space that had to be non-dimensionalized, applying the specific mathematical procedure detailed by Yamada [5] results in the Statz-deMars equations being reduced into the Yamada model as shown below:

$$\begin{aligned} \frac{dG}{dt} &= \gamma_G(A - G - GI) \\ \frac{dI}{dt} &= (G - Q - 1)I \\ \frac{dQ}{dt} &= \gamma_Q(B - Q - aQI) \end{aligned} \tag{6}$$

To formalize this isomorphism, the parameters governing the HP model can be mapped to the parameters of the laser cavity, as shown in the table below:

Yamada	**HP Equivalence**	**Physical Definition (Laser Dynamics)**
G, I, Q	x, y, z	Optical Gain, Photon Intensity, Saturable Absorber
A	R_0, K_0	Optical pump rate (analogous to intrinsic resource growth)
B	d_1	Small-signal absorption (analogous to prey mortality)
a	b_1, b_2	Absorber bleaching parameter (analogous to handling time)
γ_G, γ_Q	Scaling	Relaxation rates of the gain and absorber

Table 2: Parameter mapping demonstrating the structural isomorphism between the Hastings-Powell model and the optical Yamada model.

Note the structural similarities between (4) and (6). In the HP model, the cross-multiplication in the $f_1(x)y$ and $f_2(y)z$ terms represents the rate at which the prey consumes the resource and the rate at which the predator consumes the prey. The GI and QI terms in the Yamada model play the same structural role, representing the rate at which photons deplete the optical gain via stimulated emission, and the rate at which the absorber depletes the photons.

A central feature of this isomorphism lies in the saturation mechanics of the top-level regulator. In the Yamada model, the critical non-linearity is embedded within the aQI term, which dictates the depletion of the saturable absorber. The parameter a specifically represents the bleaching rate; that is, how quickly the absorber's atoms become fully excited and transparent to incoming light. The optical bleaching is mathematically and dynamically identical to the handling time parameter b_i found in the Holling Type II functional response of the ecological model. Similar to how a high handling time limits a predator's ability to consume prey (irrespective of prey abundance), a high bleaching rate severely limits the absorber's capacity to regulate intense photon bursts.

In both systems, this saturation acts as a critical choke point. The top regulator (the predator in the HP model and the absorber in the Yamada model) becomes overwhelmed and loses its top-down control over the system. This allows the intermediate level (the prey in the HP model and the photons in the Yamada model) to escape this regulation and undergo rapid exponential

growth, resulting in a strong transient event. This explosive growth becomes unsustainable as it rapidly depletes the foundational resource (the resource in the HP model and the optical gain in the Yamada model). The depletion forces the intermediate population to crash, allowing the top regulator to recover and thus resets the chaotic cycle.

Because these two models share mathematically identical mechanisms of accumulation, release, and saturation, their underlying phase spaces are topologically equivalent during the initial onset of instability. As their respective saturation parameters (b_1 for the HP model, a for the Yamada model) are increased, both systems lose stable steady-states and undergo a Hopf bifurcation to form limit cycles, resulting in the massive transient bursts characteristic of ecological outbreaks and Q-switched optical pulses.

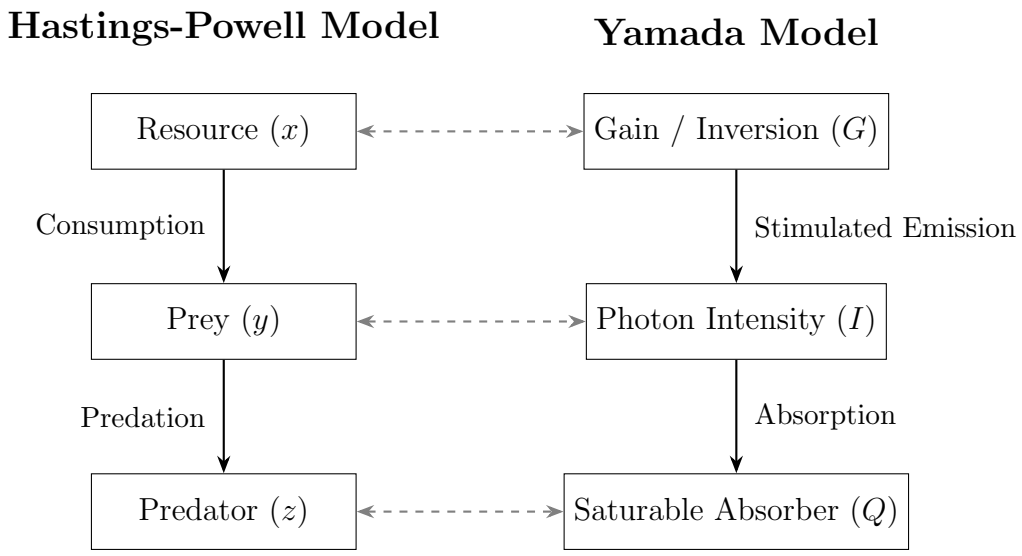


Figure 4: Diagram illustrating the structural isomorphism between the Hastings-Powell model and the Yamada model. Horizontal dashed lines indicate mathematically analogous variables.

However, while the structural isomorphism between the Hastings-Powell and Yamada models holds for their transient accumulation-and-release phases where both systems exhibit periodic attractors, their asymptotic behaviors diverge in the parameter regimes studied here. As the saturation parameter of the HP model is increased, its phase space folds and the system passes through a period-doubling cascade into a strange attractor. In the parameter regime simulated here, the baseline Yamada model shows no such folding: its pulse train remains period-1. This is an empirical observation for this regime, not a consequence of the Yamada model being purely quadratic or of it lacking dimensions; three-dimensional ODEs with only quadratic nonlinearities, such as the Lorenz and Rössler systems, can be chaotic [2]. In laser physics, chaos with a saturable absorber is commonly induced by adding delayed optical feedback; i.e., a physical setup where emitted light is reflected back into the cavity by an external mirror after a short time delay. Mathematically, this feedback transforms the Yamada model from a simple system of ordinary differential equations (ODEs) into an infinite-dimensional set of delayed differential equations (DDEs) capable of sustaining chaotic dynamics. However, exploring this highly complex DDE regime falls outside the scope of this project, which deliberately restricts its focus to the baseline ODE formulations to isolate the fundamental accumulation-and-release mechanics. Because the simulated Yamada model operates without this delay, its phase space trajectories remain bound to a predictable periodic limit cycle across the whole parameter range examined. Thus, the boundary of the structural isomorphism observed in this project is: the baseline Yamada ODE model mirrors the periodic ecological outbreak, but in the regime studied it does not reproduce the subsequent biological chaos within the HP model.

3 Equilibria and Linear Stability Analysis

To fully understand the onset of chaos in the HP model, we can first start by analyzing the system's equilibrium points. These are states where all population growth rates are zero, that is: $\frac{dx}{dt} = 0$, $\frac{dy}{dt} = 0$, $\frac{dz}{dt} = 0$. Solving these equations simultaneously yields the nullsurfaces of the system. Finding the intersections of these surfaces in the biologically feasible region ($x, y, z \geq 0$) identifies four equilibrium points as detailed below.

3.1 Identification of Equilibrium Points

1. Trivial Equilibrium (E_0):

$$E_0 = (0, 0, 0). \tag{7}$$

This represents the state of total extinction in which predator, prey, and resource populations have all become extinct.

2. Resource-Only Equilibrium (E_1):

$$E_1 = (1, 0, 0). \tag{8}$$

This represents a state where the resource population grows continuously, free from predation, until it reaches its environmental carrying capacity (which is 1 in our non-dimensionalized model).

3. The Predator-Free Coexistence Equilibrium (E_2): This state exists when the prey and resource populations coexist but the predator is absent from the system ($z = 0$). We can find the steady-state resource population x^* setting $dy/dt = 0$ and factoring out y:

$$\frac{a_1 x}{1 + b_1 x} - d_1 = 0. \tag{9}$$

Solving for x returns the critical resource threshold needed to sustain the prey:

$$x^* = \frac{d_1}{a_1 - d_1 b_1}.$$

Substituting x^* into $dx/dt = 0$ lets us solve for the steady-state prey population y^*,

$$x^*(1 - x^*) - \frac{a_1 x^*}{1 + b_1 x^*} y^* = 0 \implies y^* = \frac{x^*(1 - x^*)}{d_1}.$$

This gives us the predator-free equilibrium point,

$$E_2 = (x^*, y^*, 0). \tag{10}$$

In this state, the system often undergoes a Hopf bifurcation as the prey's mortality rate (d_1) varies, causing the system to transition from a stable spiral to limit-cycle oscillations.

4. Interior Coexistence Equilibrium (E_3): This equilibrium state is where all three species coexist; that is: $(x, y, z > 0)$. First, we can set $dz/dt = 0$ and factor out z to get the steady-state prey population $\bar{y}$,

$$\frac{a_2 y}{1 + b_2 y} - d_2 = 0 \implies \bar{y} = \frac{d_2}{a_2 - d_2 b_2}.$$

Taking this value for $\bar{y}$ and substituting it into $dx/dt = 0$ gives us the steady state resource population $\bar{x}$. This requires solving the quadratic equation

$$\bar{x}(1 - \bar{x}) = \frac{a_1 \bar{x}}{1 + b_1 \bar{x}} \bar{y} \implies b_1 \bar{x}^2 + (1 - b_1)\bar{x} + (a_1 \bar{y} - 1) = 0.$$

When $a_1\bar{y} < 1$ (as for all parameter sets used in this report, where $a_1\bar{y} = 0.625$), the constant term is negative, so the product of the two roots is negative and exactly one root is positive; we select this positive root to obtain $\bar{x}$. Finally, substituting both $\bar{x}$ and $\bar{y}$ into $dy/dt = 0$ yields the steady-state predator population $\bar{z}$,

$$\bar{z} = \frac{1}{d_2}\left(\frac{a_1\bar{x}}{1 + b_1\bar{x}}\bar{y} - d_1\bar{y}\right).$$

Thus, we have that the fully coexistent equilibrium is,

$$E_3 = (\bar{x}, \bar{y}, \bar{z}). \tag{11}$$

This is what eventually leads to chaos with respect to the Hastings-Powell model. As parameters such as the prey saturation constant shift, the equilibrium point E_3 loses stability, and further increases push the three-dimensional trajectories onto a strange attractor.

3.2 Jacobian Matrix and Local Stability

To determine the local stability of the biologically feasible equilibrium points, we must first linearize the non-linear Hastings-Powell system. We achieve this by computing the general Jacobian matrix, $J(x, y, z)$, which consists of the partial derivatives of the system's differential equations with respect to each state variable, yielding

$$J(x,y,z) = \begin{pmatrix} 1 - 2x - \frac{a_1 y}{(1+b_1 x)^2} & -\frac{a_1 x}{1+b_1 x} & 0 \\ \frac{a_1 y}{(1+b_1 x)^2} & \frac{a_1 x}{1+b_1 x} - \frac{a_2 z}{(1+b_2 y)^2} - d_1 & -\frac{a_2 y}{1+b_2 y} \\ 0 & \frac{a_2 z}{(1+b_2 y)^2} & \frac{a_2 y}{1+b_2 y} - d_2 \end{pmatrix}. \tag{12}$$

Jacobian Matrix for E_0:

Substituting $x, y, z = 0$ into Eq. 12 for the equilibrium point E_0 gives the matrix

$$J_0(0,0,0) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & -d_1 & 0 \\ 0 & 0 & -d_2 \end{pmatrix} \tag{13}$$

This is a diagonal matrix, which means the eigenvalues are simply the entries along the diagonal, giving us

$$\begin{aligned} \lambda_1 &= 1 \\ \lambda_2 &= -d_1 \\ \lambda_3 &= -d_2. \end{aligned}$$

We know that d_1 and d_2 represent mortality rates and are strictly positive. Therefore, $\lambda_2 = -d_1$ and $\lambda_3 = -d_2$ are strictly negative. Introducing prey or predators into an empty environment simply results in them dying off and their populations returning to zero. This means that this state is stable along the Prey and Predator (y and z) axes. It is also evident that $\lambda_1 = 1$ is positive, which means the system is unstable along the Resource (x) axis. Introducing even a single unit of the resource will cause its population to grow exponentially away from zero, which is what we would expect in a system with no predation.

Mathematically, this equilibrium point E_0 is classified as a saddle point, where trajectories are "pulled in" along certain axes and "pushed away" along other axes.

Jacobian Matrix for E_1:

Evaluating the matrix from Eq. 12 at the equilibrium point $E_1(1,0,0)$ yields

$$J_1(1,0,0) = \begin{pmatrix} -1 & -\frac{a_1}{1+b_1} & 0 \\ 0 & \frac{a_1}{1+b_1} - d_1 & 0 \\ 0 & 0 & -d_2 \end{pmatrix}. \tag{14}$$

This is an upper triangular matrix, which means that the eigenvalues are simply the entries along the diagonal,

$$\begin{aligned} \lambda_1 &= -1 \\ \lambda_2 &= \frac{a_1}{1+b_1} - d_1 \\ \lambda_3 &= -d_2. \end{aligned}$$

It is evident that $\lambda_1 = -1$ is negative. Since the resource is already at carrying capacity ($x = 1$), any perturbation to its population will simply result in it naturally settling back to 1. We also know that d_2 represents the mortality rate of the predator and is strictly positive. Thus, $\lambda_3 = -d_2$ is strictly negative. Introducing any predators into a system without prey will result in their population dying off and returning to 0. Thus, we can conclude that this state is stable along the Predator and Resource (z and x) axes. As for $\lambda_2 = \frac{a_1}{b_1+1} - d_1$, there are two possible cases:

- If $\frac{a_1}{b_1+1} > d_1$ (meaning $\lambda_2 > 0$), the prey can consume enough to overcome its death rate. This means the prey can successfully invade the system and grow unimpeded. In this case E_1 becomes an unstable saddle point and the system transitions toward the coexistence state E_2, discussed in further detail below.
- If $\frac{a_1}{b_1+1} < d_1$ (meaning $\lambda_2 < 0$), the prey's natural death rate is high enough such that it can not be overcome no matter how much it consumes; that is the prey will eventually die out and go extinct, which means that E_1 is a stable node.

Jacobian Matrix for E_2:

By taking the matrix from Eq. 12 and substituting $z = 0$ and $\frac{a_1 x^*}{1+b_1 x^*} = d_1$ from Eq. 9 we obtain the matrix

$$J_2(x^*, y^*, 0) = \begin{pmatrix} 1 - 2x^* - \frac{a_1 y^*}{(1+b_1 x^*)^2} & -d_1 & 0 \\ \frac{a_1 y^*}{(1+b_1 x^*)^2} & 0 & -\frac{a_2 y^*}{1+b_2 y^*} \\ 0 & 0 & \frac{a_2 y^*}{1+b_2 y^*} - d_2 \end{pmatrix}. \tag{15}$$

Since the third row only contains one non-zero entry along the diagonal, this is a block upper-triangular matrix. This means the third eigenvalue is simply the bottom-right entry of the matrix,

$$\lambda_3 = \frac{a_2 y^*}{1+b_2 y^*} - d_2$$

Here, there are two possible cases:

- If $\frac{a_2 y^*}{1+b_2 y^*} > d_2$ (meaning $\lambda_3 > 0$), the predator can consume enough to overcome its death rate. This means the predator can successfully invade the system and grow unimpeded. In this case the system is pushed away from E_2 and toward the 3-species coexistence state E_3
- If $\frac{a_2 y^*}{1+b_2 y^*} < d_2$ (meaning $\lambda_3 < 0$), the predator's natural death rate is high enough such that it can not be overcome no matter how much it consumes; that is the predator will eventually die out and go extinct, which means that E_2 is stable along the Predator (z) axis.

To determine the full stability of E_2, we must analyze the two remaining eigenvalues λ_1 and λ_2 which govern the resource - prey subsystem. These eigenvalues can be determined from the 2×2 upper-left sub-matrix of J_2 by solving for λ from the characteristic equation $\det(J_{sub} - \lambda I) = 0$, yielding

$$\det(J_{sub} - \lambda I) = \begin{vmatrix} 1 - 2x^* - \frac{a_1 y^*}{(1+b_1 x^*)^2} - \lambda & -d_1 \\ \frac{a_1 y^*}{(1+b_1 x^*)^2} & -\lambda \end{vmatrix} = 0.$$

Expanding this determinant yields a quadratic equation in terms of λ,

$$\lambda^2 - \left(1 - 2x^* - \frac{a_1 y^*}{(1 + b_1 x^*)^2}\right)\lambda + d_1 \left(\frac{a_1 y^*}{(1 + b_1 x^*)^2}\right) = 0.$$

Temporarily defining the coefficient of λ as τ and the constant term as δ, we can apply the quadratic formula to obtain the exact explicit eigenvalues for the subsystem,

$$\lambda_{1,2} = \frac{1}{2}\left(\tau \pm \sqrt{\tau^2 - 4\delta}\right),$$

where

$$\tau = 1 - 2x^* - \frac{a_1 y^*}{(1 + b_1 x^*)^2}$$
$$\delta = d_1 \left(\frac{a_1 y^*}{(1 + b_1 x^*)^2}\right).$$

According to the Routh-Hurwitz criterion for a second-degree polynomial, the roots of the characteristic equation will have strictly negative real parts, ensuring local asymptotic stability, if and only if all the coefficients of the polynomial $\lambda^2 - \tau\lambda + \delta = 0$ are positive. The leading coefficient is evidently 1, and thus we now have 2 mathematical conditions on the subsystem:

- The coefficient of λ must be positive: $-\tau > 0 \implies \tau < 0$.
- The constant term must be positive: $\delta > 0$.

Since all equilibrium populations and parameters in this ecological model are strictly positive, the second condition is inherently satisfied; that is $\delta > 0$. The stability of the resource-prey subsystem E_2 therefore depends entirely on τ. We thus apply the quadratic formula to obtain the explicit eigenvalues and thus study the oscillatory dynamics around E_2:

$$\lambda_{1,2} = \frac{1}{2}\left(\tau \pm \sqrt{\tau^2 - 4\delta}\right).$$

We have established that $\delta > 0$. Thus, the discriminant $\tau^2 - 4\delta$ frequently becomes negative as parameters vary, resulting in complex eigenvalues $\lambda_{1,2} = \alpha + i\beta$ where the real part $\alpha = \frac{\tau}{2}$ and the imaginary part $\beta = \frac{\sqrt{4\delta - \tau^2}}{2}$.

The imaginary portion $i\beta$ dictates the frequency of the oscillations between the resource and prey, while the real portion $\alpha = \frac{\tau}{2}$ dictates whether these oscillations grow or decay. If the Routh-Hurwitz condition is satisfied ($\tau < 0$), the real part is negative and the resource-prey subsystem acts as a stable spiral. However, if τ crosses zero and becomes positive, the Routh-Hurwitz condition is violated, the real part becomes positive, and the system is no longer stable. The exact threshold where $\tau = 0$ yields purely imaginary eigenvalues; that is $\lambda_{1,2} = \pm i\sqrt{\delta}$. This is known mathematically as a Hopf bifurcation, marking the point where the stable steady-state transitions into limit-cycle oscillations.

Jacobian Matrix for E_3: By taking the matrix from Eq. 12 and substituting $x = \bar{x}$, $y = \bar{y}$, and $z = \bar{z}$, and recognizing that the predator's steady-state condition dictates $\frac{a_2\bar{y}}{1+b_2\bar{y}} - d_2 = 0$, the bottom right entry of the Jacobian for E_3 evaluates to simply 0, yielding

$$J_3(\bar{x},\bar{y},\bar{z}) = \begin{pmatrix} 1 - 2\bar{x} - \frac{a_1\bar{y}}{(1+b_1\bar{x})^2} & -\frac{a_1\bar{x}}{1+b_1\bar{x}} & 0 \\ \frac{a_1\bar{y}}{(1+b_1\bar{x})^2} & \frac{a_1\bar{x}}{1+b_1\bar{x}} - \frac{a_2\bar{z}}{(1+b_2\bar{y})^2} - d_1 & -\frac{a_2\bar{y}}{1+b_2\bar{y}} \\ 0 & \frac{a_2\bar{z}}{(1+b_2\bar{y})^2} & 0 \end{pmatrix}. \tag{16}$$

Solving the characteristic equation $\det(J_3 - \lambda I) = 0$ returns a cubic polynomial as shown below,

$$\lambda^3 + A\lambda^2 + B\lambda + C = 0.$$

Rather than explicitly calculating the roots of this cubic equation, which would prove to be rather difficult, the local stability of E_3 is traditionally determined using the Routh-Hurwitz stability criterion. This theorem states that E_3 is locally stable (that is, all eigenvalues possess strictly negative real values) if and only if the following three conditions are met simultaneously:

$$A > 0, \quad C > 0, \quad AB - C > 0$$

In terms of the Jacobian matrix entries j_{kl}, these coefficients evaluate to $A = -(j_{11} + j_{22})$, $B = j_{11}j_{22} - j_{12}j_{21} - j_{23}j_{32}$, and $C = j_{11}j_{23}j_{32}$. As detailed in the original analysis by Hastings and Powell [1], the first two conditions ($A > 0$ and $C > 0$) depend heavily on the sign of j_{11}. Because the product $(j_{23}j_{32})$ is intrinsically negative, $C > 0$ requires that $j_{11} < 0$. Furthermore, $A > 0$ requires that $j_{11} + j_{22} < 0$. While j_{11} is not mathematically guaranteed to be negative across the entire theoretical parameter space, Hastings and Powell established that within the biologically realistic regimes leading to chaos, the strong self-regulating negative feedback of the resource's growth ensures $j_{11} < 0$, thereby successfully satisfying both $A > 0$ and $C > 0$.

As a result, the loss of stability of E_3 occurs when parameter variations cause the inequality $AB - C > 0$ to be violated. At this threshold, a pair of complex conjugate eigenvalues crosses the imaginary axis into the right-hand plane. The fully coexistent equilibrium E_3 loses its stability through a Hopf bifurcation and the system's trajectories are dynamically repelled from the steady state and transition into sustained limit-cycle oscillations. This marks the onset of oscillations, not yet of chaos; chaos only appears after further period-doubling bifurcations of the limit cycle. Rather than explicitly solving this algebraic inequality, this project locates the threshold numerically: for the parameters of Section 4, $AB-C$ changes sign at $b_1 \approx 2.114$, where the eigenvalues of J_3 are $\lambda \approx -0.401$ and $\lambda \approx \pm 0.114i$. As will be demonstrated numerically in Section 4, sweeping the resource saturation parameter b_1 beyond this $AB - C = 0$ boundary first produces a limit cycle and then triggers the period-doubling cascade that ultimately gives way to bounded strange attractors.

4 Computational Methods and Numerical Results

Having established the theoretical framework and the stable and unstable fixed points of the Hastings-Powell model, we now move on to the computational methods required to study its transient and chaotic dynamics. Chaotic systems are extremely difficult to simulate due to their inherent instability; thus robust computational methodologies are used with the time-series analysis to ensure accuracy. Because of the structural isomorphism established in 2.3, the computational architecture defined below in this section can be applied simultaneously to both the HP and Yamada models.

4.1 Numerical Integration and Time-Series Analysis

The continuous-time ODEs defining the Hastings-Powell system cannot be solved analytically. To approximate solutions computationally, we must employ numerical integration. However, standard fixed-step methods such as the classic 4th-order Runge-Kutta (RK4) are often insufficient for chaotic systems. In regions of phase space where trajectories diverge exponentially or curl tightly around an unstable focus, fixed step sizes accumulate truncation errors that artificially perturb the system, causing the simulation to drift off the true strange attractor.

To mitigate this, all numerical integrations were performed in Python utilizing the standard SciPy and NumPy scientific libraries [6, 7], specifically employing the adaptive step-size 8th-order Dormand-Prince (DOP853) algorithm. The architectural framework for managing and integrating these continuous-time chaotic models was adapted from the `dysts` dynamical systems library [8]. At each time step, DOP853 computes an 8th-order solution together with lower-order embedded estimates of the local error. If this error estimate exceeds a stringent predefined threshold (set to absolute and relative tolerances of 10^{-10}), the algorithm automatically shrinks the time step to preserve accuracy.

Between Q-switched pulses the photon intensity of the Yamada model falls to 10^{-5} or below, and an explicit integrator can then overshoot through zero and produce negative, unphysical intensities. The Yamada model was therefore integrated in the variable $u = \ln I$, for which $dI/dt = (G-Q-1)I$ becomes exactly $du/dt = G-Q-1$; this keeps $I = e^u > 0$ by construction.

To numerically validate the structural isomorphism, both systems were integrated using this architecture over a time-span of $t \in [400, 800]$. The HP model was initialized at $(x_0, y_0, z_0) = (0.8, 0.2, 8.0)$ with $(a_1, a_2, b_2, d_1, d_2) = (5.0, 0.1, 2.0, 0.4, 0.01)$ and $b_1 = 2.2$, which lies strictly within the period-1 limit cycle regime ($2.114 < b_1 < 2.29$; see Section 4.2), preventing the onset of deterministic chaos in order to isolate the baseline accumulation-and-release dynamics. Correspondingly, the Yamada model was initialized at $(G_0, I_0, Q_0) = (5.0, 0.1, 4.0)$ with $A = 6.5$, $B = 5.0$, $a = 4.0$ and $\gamma_G = \gamma_Q = 0.02$. Since $A > B + 1$, the non-lasing state $I = 0$ is unstable and the laser operates in the self-pulsing (Q-switching) regime, ensuring continuous optical bursting.

As shown in Figure 5, both systems exhibit mathematically analogous periodic transients. In the ecological model (top), the prey population temporarily escapes top-down regulation, resulting in periodic transient outbreaks and subsequent crashes. This behavior is mirrored by the Yamada model (bottom), where the photon intensity undergoes violent, repeating Q-switched pulses. While the visual profiles differ, (the ecological model displaying rolling asymmetrical waves and the laser model displaying extreme spikes) this disparity does not stem from physical scales, as both models are strictly non-dimensional. Rather, it is a result of their distinct mathematical non-linearities: the rational fractional form of the Holling Type II response versus the purely quadratic terms of the Yamada model, together with the slow relaxation rates $\gamma_G, \gamma_Q \ll 1$ of the laser. Despite this, both models undergo a Hopf bifurcation driven by an identical mechanism of regulator saturation followed by rapid release.

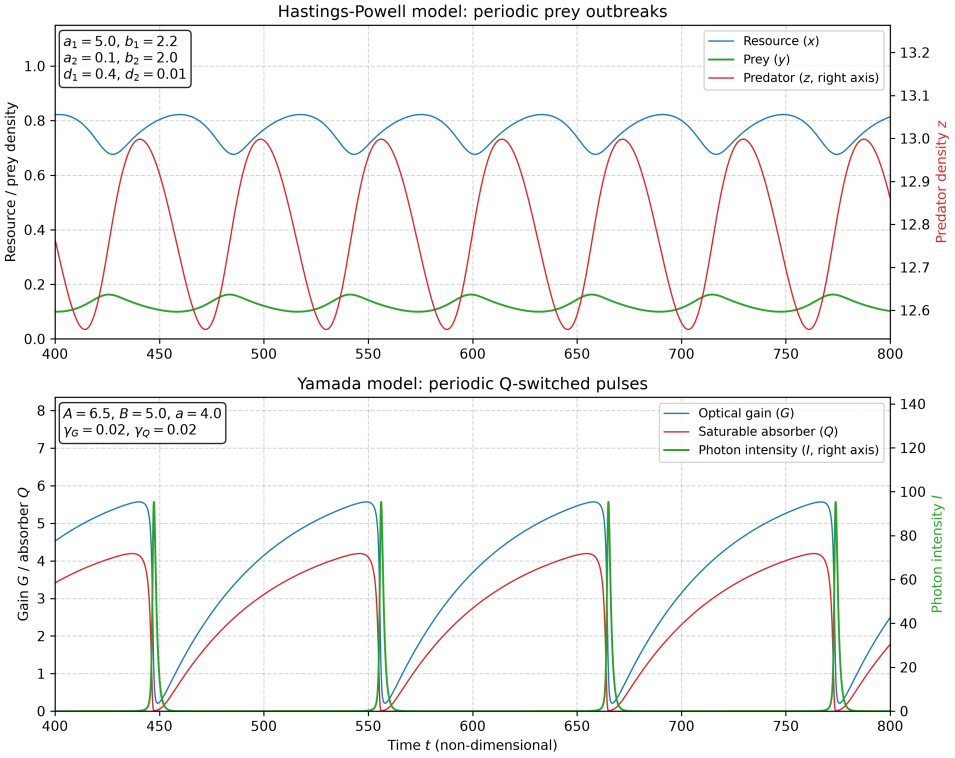


Figure 5: Time-series evolution of the Hastings-Powell (top, $b_1 = 2.2$) and Yamada (bottom, $a = 4$) models integrated using the DOP853 algorithm. The parameters are deliberately tuned such that both systems operate in a period-1 limit cycle regime. The predator z and photon intensity I are plotted against the right-hand axes. The visual similarity between the periodic ecological outbreaks (green, top) and the Q-switched optical pulses (green, bottom) illustrates the structural similarity of their respective accumulation-and-release mechanisms.

4.2 Bifurcation Analysis and Limit Cycles

Having established the fundamental accumulation-and-release mechanics in the time domain, we systematically map the transition from stable equilibria to chaos by plotting a bifurcation diagram. Based on our linear stability analysis in Section 3 of the fully coexistent equilibrium (E_3), the stability of the system hinges on the Routh-Hurwitz condition, $AB - C > 0$. Because the resource saturation constant b_1 heavily dictates the Jacobian matrix entries, it serves as a powerful bifurcation parameter. By sweeping b_1, we can force the system to violate the Routh-Hurwitz criteria, mathematically pushing the complex conjugate eigenvalues across the imaginary axis and triggering a Hopf bifurcation.

The computational algorithm for generating this diagram was executed by incrementally varying the parameters across a defined interval, integrating the system over a long time horizon ($t = 10000$ for the HP model and $t = 40000$ for the Yamada model, whose decay to steady lasing is very slow near its own Hopf point), and discarding a transient period so that only the final 5000 time units, after the trajectory has settled onto its asymptotic state, are retained. For the remaining time series, a peak-finding algorithm from the SciPy signal processing module [6] extracted the local maxima of the intermediate variables (prey population y and photon intensity I).

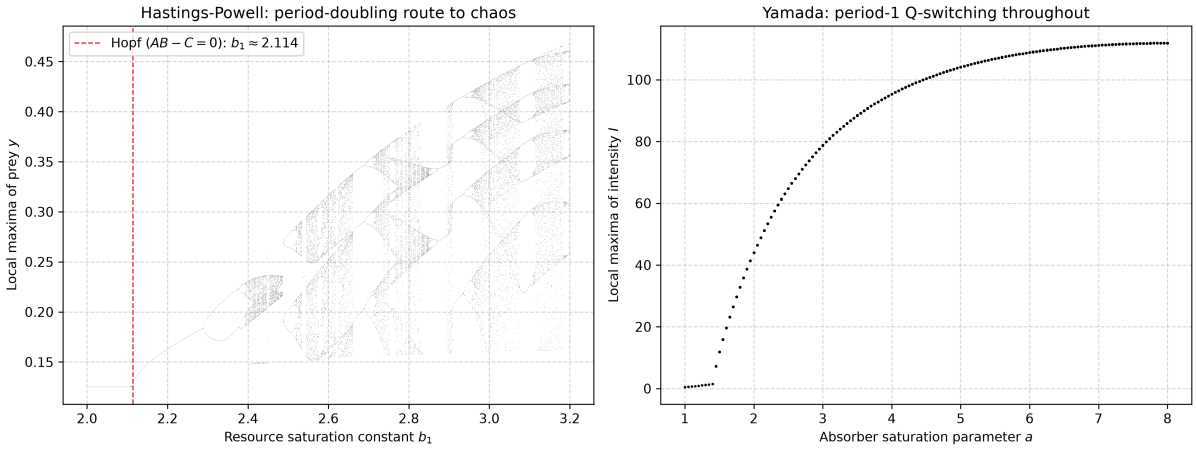


Figure 6: Bifurcation diagrams for the HP ecological system (left) and the Yamada laser model (right). The ecological system loses its stable equilibrium at the Hopf point $b_1 \approx 2.114$ (dashed line) and then undergoes a period-doubling cascade into a chaotic strange attractor as the resource saturation constant (b_1) increases. In contrast, once it starts pulsing ($a \gtrsim 1.4$), the optical system maintains a stable period-1 limit cycle whose amplitude grows as the absorber bleaching parameter (a) increases.

Figure 6 illustrates the stark contrast in the asymptotic limits of the two systems. In the HP model (left), for $b_1 < 2.114$ the equilibrium E_3 is stable and the local maxima collapse onto the single equilibrium value $\bar{y} = 0.125$. At $b_1 \approx 2.114$ the Routh-Hurwitz quantity $AB - C$ changes sign, E_3 undergoes a Hopf bifurcation, and the system settles into a stable period-1 limit cycle. As b_1 increases further, the system undergoes a series of period-doubling bifurcations (numerically at $b_1 \approx 2.29$, 2.38 and 2.39), a universal mathematical route to chaos known as a Feigenbaum cascade [9]. The local maxima extract two, then four, then eight discrete points before rapidly dissolving into a dense strange attractor, interrupted by periodic windows.

Conversely, the Yamada model (right) exhibits a fundamentally different result. For small a the laser emits continuously (a single, steady intensity value); at $a \approx 1.4$ this steady state undergoes its own Hopf bifurcation and the laser starts pulsing. While the Yamada model is capable of exhibiting chaos under different specific configurations (such as with the addition of external modulation or delayed optical feedback), within this parameter regime no period-doubling occurs. As a result, the local maxima of the photon intensity remain bound to a single highly predictable period-1 limit cycle across the whole range $1.4 \lesssim a \leq 8$ examined.

4.3 Sensitive Dependence on Initial Conditions (SDIC)

A defining hallmark of a chaotic strange attractor is its sensitive dependence on initial conditions (SDIC)[3]. To quantify this numerically, we simulate two systems simultaneously: a baseline trajectory $\mathbf{u_1}(t)$ initialized at $\mathbf{u_0}$ and a perturbed trajectory $\mathbf{u_2}(t)$ initialized at $\mathbf{u_0} + \delta\mathbf{u_0}$ where $\delta\mathbf{u_0}$ represents a small perturbation on the order of 10^{-5}. Here $\mathbf{u_0}$ is a point already on the attractor ($b_1 = 3.0$). The Euclidean distance between these two states in phase space is defined as:

$$D(t) = ||\mathbf{u}_1(t) - \mathbf{u}_2(t)|| = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2 + (z_1 - z_2)^2} \tag{17}$$

In a chaotic system, this separation distance grows exponentially according to the relationship $D(t) \approx ||\delta||e^{\lambda t}$. As shown in the upper panel of Figure 7, plotting $D(t)$ on a logarithmic scale visually confirms this exponential divergence, generating a (noisy) linear upward trend before

eventually saturating the physical boundaries of the attractor. This confirms the presence of a positive global Lyapunov exponent. Computed from the tangent (variational) equations over $t = 20000$, and confirmed independently with the two-trajectory renormalisation method of Benettin, its value is $\lambda_1 \approx 0.0067$, i.e. errors grow by a factor e roughly every 150 time units. Long-term forecasting of the ecological state is therefore fundamentally impossible.

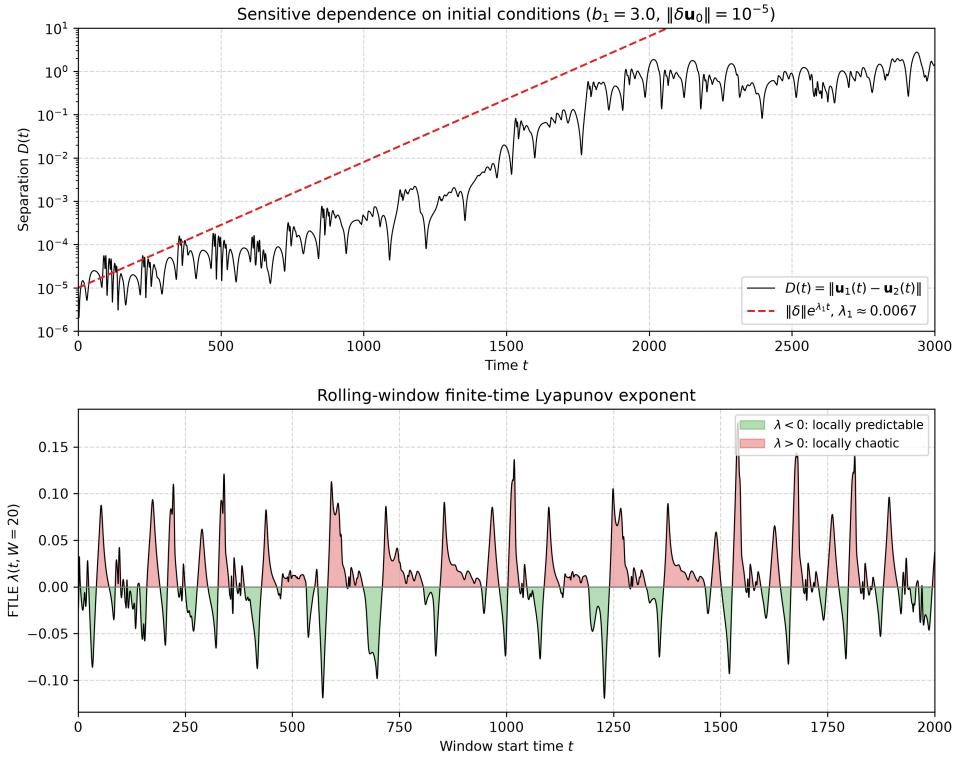


Figure 7: Exponential divergence of two closely initialized trajectories on a logarithmic scale, confirming SDIC; the dashed line has slope $\lambda_1 \approx 0.0067$ (Top). The rolling-window FTLE, $W = 20$ (Bottom). Green shaded regions ($\lambda < 0$) represent transient periods of predictability where trajectories converge, while red regions ($\lambda > 0$) indicate locally chaotic phases.

However, an infinite-time average smooths over specific short-lived events that characterize an ecosystem's outbreaks and crashes. To measure localized behavior during transient phases, a Finite Time Lyapunov Exponent (FTLE) was evaluated over a rolling time window of $W = 20$. The algorithmic implementation for tracking this localized trajectory divergence was adapted from established open-source chaos analysis frameworks [10, 11]. A tangent vector $\delta\mathbf{u}$ is evolved with the linearised flow $\dot{\delta\mathbf{u}} = J(\mathbf{u}(t))\,\delta\mathbf{u}$ (periodically renormalised to avoid overflow), so that after an initial transient it is aligned with the most unstable direction. The FTLE at time t is defined as

$$\lambda(t, W) = \frac{1}{W} \ln \left(\frac{||\delta\mathbf{u}(t + W)||}{||\delta\mathbf{u}(t)||} \right). \tag{18}$$

The lower panel of Figure 7 reveals that chaotic unpredictability is not uniformly distributed. The system frequently drops below the $\lambda = 0$ threshold into phases of transient stability (highlighted in green); about 40% of all windows have $\lambda < 0$, even though the long-run average is positive. During these brief periods, the near-future state of the ecosystem can be reliably forecasted, provided one has sufficient precision/computing power.

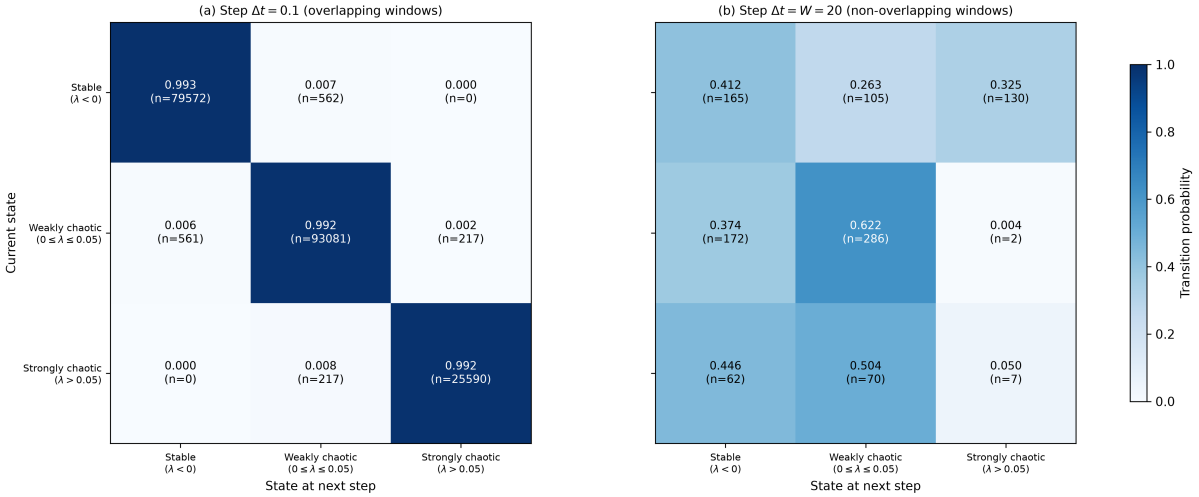


Figure 8: Transition probabilities between discretized FTLE states. (a) Consecutive FTLE values sampled every $\Delta t = 0.1$: these windows overlap almost completely, so the matrix is dominated by the diagonal and direct "Stable" $\leftrightarrow$ "Strongly chaotic" jumps are impossible by continuity. (b) Non-overlapping windows ($\Delta t = W = 20$): direct jumps between "Stable" and "Strongly chaotic" are common.

By discretizing the continuous FTLE spectrum into distinct predictive states, we can model the ecological transients as a Markov chain. As shown in Figure 8, the FTLE values over an extended integration period were categorized into three operational states: a "Stable" state ($\lambda < 0$), a "Weakly Chaotic" state ($0 \leq \lambda \leq 0.05$), and a "Strongly Chaotic" state ($\lambda > 0.05$), characterizing the explosive phase of an outbreak. When consecutive FTLE values are taken every $\Delta t = 0.1$ (panel a), the resulting transition matrix shows strong state inertia (probabilities ≈ 0.99 along the main diagonal) and no direct transitions between the "Stable" and "Strongly Chaotic" states. This, however, is a consequence of the sampling rather than of the dynamics: $\lambda(t, W)$ is a continuous function of t, and two windows that share 99.5% of their trajectory cannot differ by more than a small amount, so the FTLE must pass through the intermediate band. When the transition matrix is instead built from non-overlapping windows (panel b), each step corresponds to a fresh stretch of trajectory, and the "Stable" state is followed directly by "Strong Chaos" in about one third of cases ($p \approx 0.33$), while "Strong Chaos" is almost never followed by itself ($p \approx 0.05$). On the time scale of one window, the system therefore does not have to pass through a weakly chaotic warning phase. The discrete state-machine view nevertheless provides a computational framework for anticipating outbreaks/crashes based on historical transition probabilities, provided the sampling interval is chosen to match the forecasting horizon of interest.

4.4 Attractor Topology and Poincaré Sections

To fully reconcile the transient isomorphism and subsequent divergence of the models mathematically, we analyze the three-dimensional geometry of their phase spaces. Figure 9 visually confirms the isomorphism prior to the onset of the period-doubling cascade. Both the ecological model ($b_1 = 2.2$) and the laser model ($a = 4.0$) resolve into simple, closed, one-dimensional loops, bound to stable period-1 limit cycles.

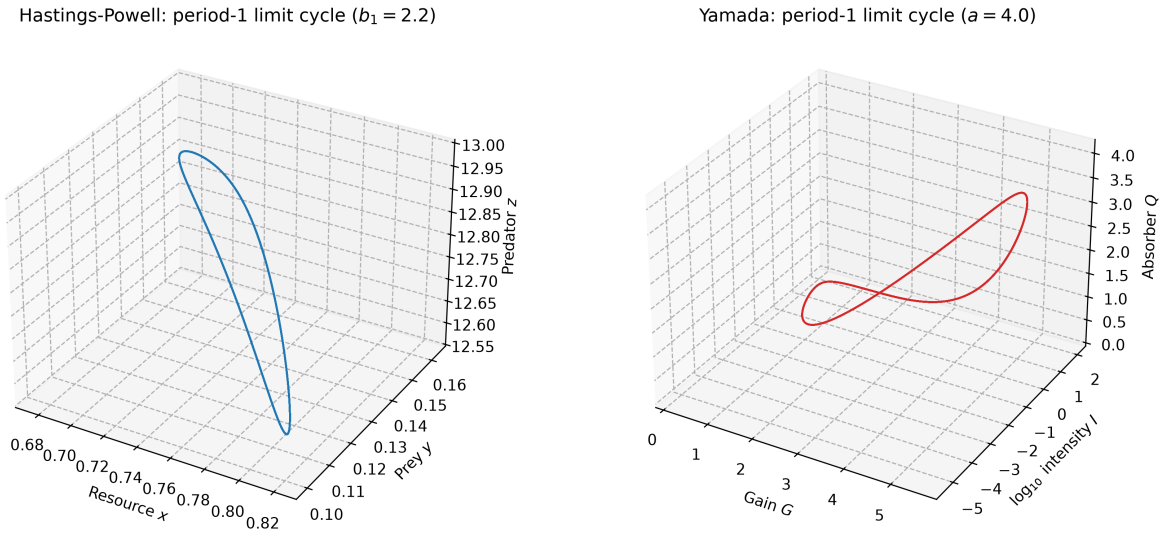


Figure 9: Phase space topologies of the Hastings-Powell model (left, $b_1 = 2.2$) and Yamada model (right, $a = 4$; intensity on a logarithmic axis) operating in their periodic regimes. Both systems form simple closed 1D manifolds.

However, as the resource saturation constant increases to $b_1 = 3.0$, the ecological model's phase space undergoes a profound geometric transformation into an infinitely layered "teacup" structure (Figure 10, left). To verify that this dense web of trajectories is driven by deterministic chaos rather than stochastic noise, a transverse Poincaré section was generated. By defining a two-dimensional transverse hyperplane Σ at a constant predator population $z = z^* = 9.0$, we restrict the recorded intersections to a unidirectional piercing ($dz/dt < 0$):

$$\Sigma = \{(x, y, z) \in \mathbb{R}^3 | z = z^*\}. \tag{19}$$

The crossings were located to integrator precision using event detection rather than by interpolating between stored samples.

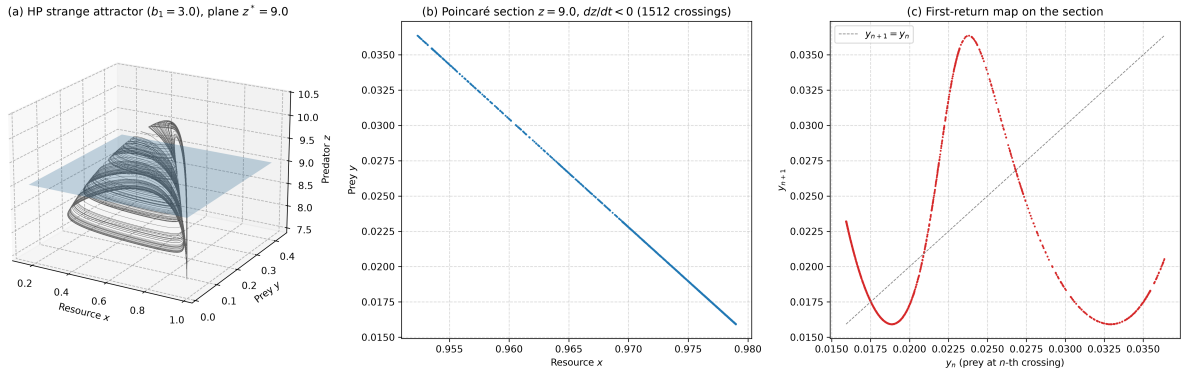


Figure 10: (a) The chaotic strange attractor of the Hastings-Powell model ($b_1 = 3.0$), intersected by a horizontal transverse hyperplane ($z = z^* = 9.0$). (b) The resulting Poincaré section mapped from unidirectional downward trajectory intersections. (c) The first-return map $y_n \mapsto y_{n+1}$ of successive crossings.

Rather than filling the 2D plane with random scatter, the intersections mapped in panel (b) of Figure 10 collapse onto a thin, essentially one-dimensional curve. This reflects the strong dissipation of the flow: the sum of the Lyapunov exponents is ≈ -0.46, so volumes in phase

space contract rapidly and the attractor is very nearly a two-dimensional sheet. The folding is revealed by the first-return map in panel (c): plotting the prey density at each crossing against its value at the next crossing gives a smooth, single-valued, multi-humped curve. A single-valued map of this kind is the signature of deterministic dynamics (noise would produce a cloud), and its non-monotonic, folded shape is the stretching-and-folding mechanism that generates chaos. This maps the boundary of the isomorphism observed here: while both models utilize similar saturation mechanisms to trigger a transient burst, the Yamada equations at the parameters studied show no such folding, and no strange attractor appears.

4.5 Correlation Dimension and Fractal Geometry

Unlike stable limit cycles (whose trajectories resolve to a 1D line), chaotic trajectories trace out a complex topology that possesses a fractional dimension. To quantify this numerically, the correlation dimension D_2 was calculated using the Grassberger-Procaccia algorithm, with implementation adapted from the `nolds` non-linear dynamics package [10]. Given a time series of N points in our 3D space, $\mathbf{u}_i = (x_i, y_i, z_i)$, we calculate the spatial distance between all possible pairs of points. We then define the correlation sum $C(r)$, which represents the probability that any two points on the attractor are separated by a distance less than r:

$$C(r) = \lim_{N\to\infty} \frac{2}{N(N-1)} \sum_{i=1}^{N} \sum_{j=i+1}^{N} H(r - \|\mathbf{u}_i - \mathbf{u}_j\|), \tag{20}$$

where H is the Heaviside step function. For a fractal object, the correlation sum scales as a power law with respect to the radius r. Therefore, the correlation dimension is extracted by taking the limit of the logarithmic derivative:

$$D_2 = \lim_{r\to 0} \frac{\log C(r)}{\log r}. \tag{21}$$

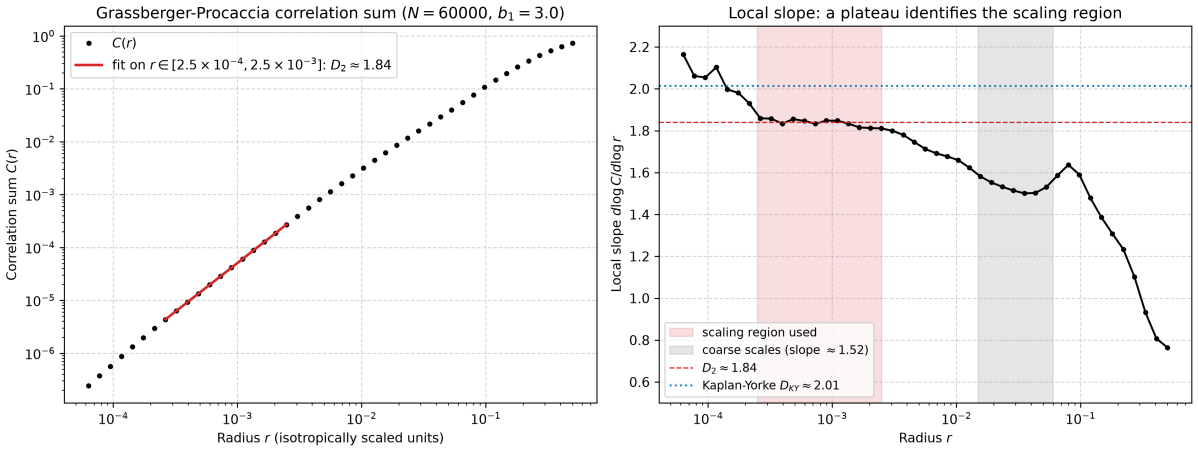


Figure 11: Left: log-log plot of the correlation sum $C(r)$ against the spatial radius r ($N = 60000$ points, $b_1 = 3.0$). Right: the local slope $d\log C/d\log r$. The plateau (shaded red) gives a fractional correlation dimension of $D_2 \approx 1.84$; fitting at coarser scales (shaded grey) would underestimate it as ≈ 1.5. The Kaplan-Yorke dimension $D_{KY} \approx 2.01$ is shown for comparison.

To execute this, the native axes were isotropically scaled by the maximum global range (a single common factor, which leaves the dimension unchanged) so that distances are measured on a unit scale. Furthermore, the time-series data was temporally decoupled by retaining only one

point every 5 time units to prevent the Theiler anomaly [12], a phenomenon where temporally adjacent points in an ODE integration artificially inflate the spatial correlation. The correlation sum $C(r)$ was then computed rapidly using the KDTree spatial indexing algorithm provided by the SciPy library [6].

As shown in Figure 11, plotting $\log_{10}(C(r))$ against $\log_{10}(r)$ reveals a distinct linear scaling region, which is identified objectively as the range of r over which the local slope $d\log C/d\log r$ is flat ($2.5\times10^{-4} \lesssim r \lesssim 2.5\times10^{-3}$). A linear regression applied to this region yields a slope of $D_2 \approx 1.84$. At coarser scales ($r \sim 0.02$–0.05) the slope dips to ≈ 1.5 because the large-scale folds of the attractor dominate, so a fit there underestimates D_2. As an independent check, the Kaplan-Yorke dimension computed from the Lyapunov spectrum $(\lambda_1, \lambda_2, \lambda_3) \approx (0.0067, 0, -0.463)$ is $D_{KY} = 2 + \lambda_1/|\lambda_3| \approx 2.01$; the correlation dimension is expected to lie at or below D_{KY}, consistent with the estimate obtained. This non-integer fractional dimension serves as numerical evidence for the boundary of the structural isomorphism: the biological ecosystem's non-linear saturation allows it to cross the threshold from 1D periodic geometry into fractal chaos.

5 Conclusions and Future Work

This project investigated the transient dynamics of the canonical Hastings-Powell trophic model by exploring a structural isomorphism with the Yamada model from laser physics. By drawing a mathematical analogy between the handling time of a biological predator and the optical bleaching of a saturable absorber, it was demonstrated that both systems rely on similar mechanisms of accumulation, saturation, and rapid release to trigger bursting phenomena. Numerical integration of both models in the time domain confirmed that prior to the onset of chaos, the ecological outbreaks and optical Q-switched pulses are topologically equivalent, each resolving to a stable period-1 limit cycle in 3D phase space.

However, the geometric mapping of these attractors highlighted a boundary to this isomorphism. Bifurcation analysis showed that the HP equilibrium loses stability through a Hopf bifurcation at $b_1 \approx 2.114$, exactly where the Routh-Hurwitz quantity $AB - C$ changes sign, and that the resulting limit cycle then passes through a Feigenbaum period-doubling cascade into a strange attractor. The Poincaré section and its first-return map revealed the stretching-and-folding structure of this attractor, and the Grassberger-Procaccia algorithm confirmed its chaotic topology, yielding a non-integer correlation dimension of $D_2 \approx 1.84$ (with Kaplan-Yorke dimension ≈ 2.01 and largest Lyapunov exponent $\lambda_1 \approx 0.0067$). Conversely, over the parameter range examined, the Yamada model showed no period-doubling and remained on a period-1 limit cycle. This is an observation for this regime rather than a structural impossibility, since three-dimensional quadratic systems can be chaotic.

Furthermore, this analysis demonstrated that the extreme unpredictability of chaotic ecological outbreaks is not temporally uniform. By computing the Finite Time Lyapunov Exponent (FTLE) over a rolling window, distinct phases of localized transient stability were isolated, covering roughly 40% of the time. By discretizing these phases, a Markov transition matrix was generated. With non-overlapping windows this matrix shows that the system frequently passes directly from locally stable to strongly chaotic behaviour, so a weakly chaotic phase cannot be relied upon as an early warning on that time scale; the apparent necessity of such an intermediate phase in finely sampled data is an artefact of overlapping windows. Future work on this subject could focus on applying this probabilistic forecasting framework, with the sampling interval matched to the forecast horizon, to empirical ecological data, such as real-world time series of aquatic algal blooms. Additionally, investigating modified optical models that incorporate delayed feedback mechanisms could establish a fuller isomorphism capable of mirroring both the periodic and chaotic limits of the Hastings-Powell ecosystem.

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